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Detecting Trajectory of Targeted Attack by Hidden Markov Model

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Abstract

The proliferation of the internet has accelerated information transmission, greatly facilitating everyday life. However, with the growth of digitalization, cyber attacks have become increasingly prevalent, and information security issues have intensified. Modern cyber attack methods are rapidly evolving, moving from random attacks to precise, targeted assaults on government agencies and enterprises, with Advanced Persistent Threats (APTs) being a prominent example. These targeted attacks often infiltrate systems undetected, lying dormant until the opportune moment to strike, frequently causing substantial damage before detection.

This study, based on real-world cyber attack data, identifies that attack behaviors often exhibit observable patterns before initiation. Therefore, the study employs the Elastic Net model to isolate critical attack characteristics, further analyzing and predicting attack behavior using a Hidden Markov Model (HMM). The HMM, grounded in sequence behavior, is capable of managing randomness and uncertainty, enabling early detection of latent attack actions before significant damage occurs.

Experimental results show that this approach significantly enhances the precision of network security monitoring and effectively reduces the potential losses caused by attacks. In an environment of increasingly complex cyber threats, this study offers a combined theoretical and practical solution to strengthen real-time warnings and bolster cybersecurity defenses.

Keywords: cyber attack, Advanced Persistent Threat (APT), Elastic Net, Hidden Markov Model (HMM), information security monitoring

1. Introduction

With the widespread adoption of the internet, cyber attacks have become increasingly frequent, organized, and systematic, making it challenging for administrators to trace attack trajectories. In recent years, the rapid development of Industrial Internet of Things (IIoT) technology has introduced a large number of connected devices and applications, intensifying security threats in industrial networks [1]. IIoT networks frequently encounter various types of cyber attacks, especially Distributed Denial of Service (DDoS) attacks, such as HTTP Flood, ICMP Flood, TCP SYN Flood, and UDP Flood [2, 3], all of which pose serious threats to

network stability and security [4]. Attackers may even combine DDoS with Advanced Persistent Threat (APT) tactics, using diversionary tactics to mask their intentions and strengthen the effectiveness of their intrusions [5].

Traditional detection methods, which often rely on predefined sequential attack steps, are no longer adequate in today's complex cyber threat landscape [6]. Attackers frequently alter their attack sequences, rendering fixed-order detection methods prone to error and misjudgment. To address this, more advanced detection and defense strategies are important. Given the high volume of data and the rapid evolution of attack techniques, current methods often fall short in terms of both speed and accuracy, necessitating more effective approaches for enhancing cybersecurity.

In response to these challenges, we initially apply the Elastic Net model to identify attributes associated with targeted attacks. Elastic Net combines Lasso (L1 Regularization) and Ridge (L2 Regularization) techniques: Lasso selects a few important features by inducing sparsity, while Ridge balances the coefficients of all features to prevent overfitting. Thus, Elastic Net is able to retain features highly relevant to attack behaviors while addressing inter-feature collinearity. This technique is particularly suitable for targeted attacks (such as APTs), which often require analyzing a large number of behavior features with significant inter-correlations.

To support this analysis, we collect data related to targeted attacks from multiple network log sources, such as IDS, IPS, WAF, and traffic records, and construct a feature matrix based on time, traffic, and operations. Each sample is labeled as either "normal" or "attack behavior," forming a feature matrix X and corresponding target vector y , where y represents targeted attack labels. All possible network behaviors, such as connection counts, IP changes, scans, and abnormal logins, are converted into numerical feature variables. Given the multi-dimensional and highly correlated nature of targeted attack behaviors, these features form a high-dimensional matrix. The Elastic Net model is then applied to allocate feature weights in the feature space, penalizing less important features by compressing their coefficients and ultimately retaining only the most representative attack characteristics. For example, retained features may include connection counts with malicious IPs, suspicious file operations, or abnormal network traffic within specific time frames, all of which help identify potential attack patterns. An unusual spike in network traffic, for instance, may signal an impending DDoS attack.

Given that the data in this study is based on real attack logs, historical data analysis enables early-stage identification of potential threats by examining the distribution of attack and non-attack behaviors and applying statistical methods to quantify risk thresholds. When system traffic, for instance, reaches a medium or high-risk threshold, a warning mechanism is triggered, alerting security personnel to intervene and prevent further escalation of the attack. This approach significantly improves network security and reduces potential damage.

Additionally, we employ the Hidden Markov Model (HMM) to analyze targeted attack behaviors. HMM is a statistical model that describes observed event sequences influenced by

hidden state sequences, with each state representing different stages of an attack process. The transition probabilities between these hidden states model the attacker's behavior pattern. In this study, the attack process is decomposed into a series of temporal stages, with each stage corresponding to a hidden state. Transition probabilities between hidden states in the HMM represent the likelihood of an attacker moving between different stages. By analyzing these transition probabilities, we can identify typical patterns in the attack process and predict potential future attack behaviors. Network traffic data is input into the HMM to identify behavior patterns associated with targeted attacks, allowing for the detection of ongoing attacks and proactive prediction and prevention of future attacks based on known attack patterns, effectively enhancing detection and prevention capabilities.

Our methodology highlights the use of HMM to analyze attack steps in a targeted attack. In the experimental phase, we validated the proposed method's effectiveness and feasibility, demonstrating its capability to address various rapidly evolving attack techniques. The primary contribution of this research is its combined theoretical and practical solution for improving network security. These contributions comprehensively outline the study's research background, motivation, methodology, objectives, and significance, with a particular emphasis on defenses against targeted attacks. This innovative approach illustrates significant value in advancing network security.

2.Related Works

This chapter introduces previous studies related to targeted attacks, the Elastic Net model, and the Hidden Markov Model (HMM). Targeted attacks are sophisticated, long-term attacks with a persistent, latent nature. The Elastic Net model allows for feature selection while retaining features highly relevant to attack behaviors and handling collinearity among features. The Hidden Markov Model is used to detect state changes in a sequence of observed events and predict attacker intentions. Through these methods, this study aims to identify and predict potential attack activities early, allowing for preventive measures to be taken before attacks occur.

2.1 Targeted Attack

Targeted attacks are highly sophisticated and advanced attack methods characterized by long-term persistence and latent infiltration. These attacks are usually directed at specific targets, carefully planned, and executed in multiple steps to achieve the attacker's goals of data extraction or system disruption. Scholars and institutions have various approaches to dividing the phases of targeted attacks. Sood categorizes the attack phases into three primary stages: intelligence gathering, threat modeling, and attacking and exploiting targets [7]. The 2021 MITRE Engenuity ATT&CK Evaluations report simulated real APT techniques and procedures used by hacker groups Carbanak and FIN7, covering 174 attack steps. The main phases include breaching defenses, maintaining internal presence, gathering intelligence, privilege escalation, lateral movement, and executing the attack to complete the objective [8, 9].

In all targeted attacks, intelligence gathering is an essential first step. Hackers collect information related to the target, including the organization's environment, department backgrounds, partners, and system configurations. They create a list of potential targets and analyze this information to narrow down their attack objectives. After completing the intelligence-gathering phase, hackers proceed to threat modeling, where they formulate an attack strategy based on the collected information and use penetration testing tools to identify vulnerabilities in the target. The goal of this phase is to identify and analyze potential attack vectors to ensure the attack's success. Hackers will escalate privileges and simulate further attacks to verify the feasibility and effectiveness of their plan, confirming the target system's weaknesses to ensure a successful attack[10].

In the attacking and exploitation phase, hackers execute the actual attack, employing tactics such as phishing emails, malware, and social engineering. Upon successfully breaching the target, they gain initial access and continue operations. Finally, they establish a persistent control channel within the target system to maintain long-term access and monitoring.

Targeted attacks are highly complex and meticulous, relying on thorough planning and execution. By understanding the stages and processes of these attacks, this study can better prevent and respond to such threats, thereby safeguarding the system and data security in this research.

2.2 Elastic Net

Elastic Net is a regularization regression method used in statistics and machine learning, combining the properties of Lasso and Ridge regression. It was developed to address issues of multicollinearity and variable selection in high-dimensional data analysis, particularly suitable for scenarios where the number of variables exceeds the number of observations, such as gene expression data, bioinformatics, and financial risk assessment [11].

In linear regression analysis, when multicollinearity (high correlation among variables) or a smaller sample size relative to the number of variables exists, ordinary least squares (OLS) often yields unstable estimates. To address these challenges, regularization methods like Ridge (L2 regularization) and Lasso (L1 regularization) were introduced. However, both have limitations: Ridge regularization does not perform variable selection, as it cannot reduce coefficients to zero; Lasso regularization, on the other hand, can randomly select one of the highly correlated variables, leading to model instability. To overcome these issues, Elastic Net combines the strengths of both Lasso and Ridge, enabling variable selection and handling multicollinearity[12, 13].

The objective function for Ridge regularization (L2) minimizes the loss function plus a penalty term on the squared sum of all coefficients:

$$\underset{\beta}{\text{minimize}} \left(\sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2 + \lambda \|\beta\|_2^2 \right)$$

where β is the regression coefficient vector and λ is the regularization parameter that controls the strength of the penalty term. $\|\beta\|_2^2$ denotes the sum of squares of all coefficients. Ridge regularization prevents overfitting by penalizing large coefficients, thus reducing model complexity and handling multicollinearity. However, it cannot perform variable selection as it only reduces coefficients toward zero rather than exactly zero.

Lasso regularization (L1) minimizes the loss function plus a penalty on the sum of absolute values of coefficients:

$$\underset{\beta}{\text{minimize}} \left(\sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2 + \lambda \|\beta\|_1 \right)$$

where $\|\beta\|_1$ represents the sum of absolute values of all coefficients. Lasso can shrink the coefficients of unimportant variables to zero, achieving variable selection and yielding a simpler, more interpretable model. However, Lasso has limitations when dealing with highly correlated variables, as it may randomly select only one, causing instability. Moreover, when the number of correlated variables exceeds the sample size, Lasso can select at most as many variables as the number of samples.

The Elastic Net objective function combines the two methods above:

$$\underset{\beta}{\text{minimize}} \left(\sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1}{2} (1 - \alpha) \|\beta\|_2^2 \right) \right)$$

where α is the balancing parameter between 0 and 1. When $\alpha = 1$, Elastic Net is

equivalent to Lasso; when $\alpha = 0$, it is equivalent to Ridge. This combination allows Elastic Net to perform variable selection while handling multicollinearity. The loss function (sum of squared residuals) measures the difference between predicted and actual values, and the regularization term combines L1 and L2 penalties, with λ controlling the regularization strength and α adjusting the proportion of L1 and L2 penalties. Geometrically, Elastic Net's constraint region lies between the diamond shape of Lasso and the circular shape of Ridge, forming a flexible shape that enables Elastic Net to inherit properties of both Lasso and Ridge.

In terms of parameter selection and model training, appropriate values of λ and α are chosen using cross-validation methods to evaluate model performance for different λ and α combinations and select the parameter combination that minimizes prediction error. Grid search may also be used, defining a set of values for λ and α in advance, training, and evaluating the model for each parameter combination. Before applying Elastic Net, data standardization (mean of 0, variance of 1) is necessary to prevent the influence of different variable scales on regularization. Model evaluation metrics typically include mean squared error (MSE) and the coefficient of determination (R^2) to measure model performance and data interpretability.

Elastic Net's advantages include performing variable selection, handling multicollinearity, providing high model stability, and being suitable for high-dimensional data. However, its drawbacks include higher computational complexity due to tuning both λ and α , increasing computational load, and potentially more complex interpretation. In result interpretation, regression coefficient magnitude and direction are analyzed to understand each variable's impact on the target, and residual distribution is checked to assess model assumptions.

Elastic Net has extensive applications in various fields. In gene expression analysis, where gene data typically contains a large number of features but a limited sample size, Elastic Net is used to select gene groups related to specific diseases, improving both the interpretability and predictive performance of the model. In financial risk assessment, where numerous financial indicators may be highly correlated, Elastic Net considers multiple related financial indicators to establish a stable risk assessment model. In signal processing and image analysis, where there are large numbers of features with significant correlations, Elastic Net effectively selects relevant features from a vast pool, enhancing model performance. For instance, Wen proposed using four regression models: Ordinary Least Squares, Ridge Regression, Lasso, and Elastic Net, to train data from different domains and create a system called Nopt that can resist data poisoning in cyber attack scenarios [14].

2.3 Empirical Risk

Empirical Risk is a core concept in machine learning and statistics, playing a crucial role in supervised learning[15]. It primarily measures the prediction error of a model on the training data, specifically the average error of classification or regression on the training set. Minimizing empirical risk forms the basis of many machine learning algorithms, as it directly affects the model's learning effectiveness and generalization ability.

Given a set of training data $\{(x^t, y^t)\}_{t=1}^n$, where x^t is the input feature vector, y^t is the corresponding label, and n is the sample size, the model's parameters are denoted by θ . The

loss function $L(\theta, x^t, y^t)$ is used to evaluate the difference between the model's prediction on sample x^t and the true label y^t . The empirical risk $R_n(\theta)$ is defined as:

$$R_n(\theta) = \frac{1}{n} \sum_{t=1}^n L(\theta, x^t, y^t)$$

This formula represents the average value of the loss function across all samples, indicating the model's average prediction error on the training set.

The choice of loss function depends on the type of problem. For classification, commonly used loss functions include Cross-Entropy Loss and Log Loss. For regression problems, Mean Squared Error (MSE) and Mean Absolute Error (MAE) are frequently used. These loss functions aim to quantify the deviation between the model predictions and actual results, guiding model optimization.

Empirical Risk Minimization (ERM) is a fundamental strategy in training machine learning models, aiming to find a set of model parameters θ^* that minimize the empirical risk $R_n(\theta)$:

$$\theta^* = \arg \min_{\theta} R_n(\theta)$$

By minimizing empirical risk, the model achieves optimal predictive performance on the training data. However, solely pursuing empirical risk minimization may lead to overfitting, as the model might overly adapt to noise or anomalies in the training data, resulting in poor performance on new data.

To prevent overfitting, a regularization term is typically added to the empirical risk to form Structural Risk. Structural Risk accounts for model complexity by including a penalty value for the model parameters, such as L1 or L2 regularization. Structural Risk is defined as:

$$R_{\text{struct}}(\theta) = R_n(\theta) + \lambda \Omega(\theta)$$

where λ is the regularization parameter that controls the influence of the regularization term, and $\Omega(\theta)$ is the regularization term used to measure the model's complexity. This approach balances the fit and complexity of the model, enhancing its generalization ability.

In the application of network attack detection in this study, a suitable loss function will be defined to evaluate the model's predictive performance by measuring the difference between the model's predictions and true labels. Additionally, the empirical risk on the training data will be calculated as the objective function for model optimization. Finally, the model's generalization ability will be assessed using a test dataset to ensure effective detection of unknown network attacks. By combining the empirical risk minimization strategy with the Hidden Markov Model's powerful sequence modeling capabilities, the accuracy and reliability of attack detection can be significantly improved.

2.4 Hidden Markov Model

The Hidden Markov Model (HMM), initially published by L. E. Baum and colleagues in a series of statistical papers, has since proven valuable in fields such as speech recognition and

bioinformatics. HMM provides a well-established method for detecting state changes in a sequence of observed events over time [16,17,18]. It consists of two main parts: a hidden Markov chain and an observed sequence. The former is not directly observable but influences the latter.

In this study, given the model and the observed sequence, HMM is used to evaluate the problem. The initial values of the HMM model are defined as $\lambda = (A, B, \pi)$, where A is the state transition probability matrix, B is the observation probability matrix, and π is the initial state probability distribution. The probability of an observed sequence $O = \{o_1, o_2, \dots, o_t\}$ can be calculated using the Forward Algorithm:

$$\alpha_t(j) = \sum_{i=1}^N [\alpha_{t-1}(i) a_{ij}] \times b_j(o_t)$$

The Forward Algorithm is commonly used to efficiently calculate the probability of an observed sequence within the HMM model. Here, $\alpha_t(j)$ represents the probability of being in state j at time t while observing $o_1, o_2, \dots, o_t$.

To calculate the hidden state sequence for a given observed sequence, algorithms such as the Viterbi Algorithm and Baum-Welch Algorithm are commonly used. In this study, for known output sequences, the Baum-Welch Algorithm is used to find the most likely state transitions and output probabilities. Its recursive formula is as follows:

$$\begin{aligned} \xi_t(i, j) &= P(s_t = i, s_{t+1} = j | O, \lambda) \\ &= \frac{P(s_t = i, s_{t+1} = j, o | \lambda)}{P(o | \lambda)} = \frac{\alpha_t(i) a_{ij} \cdot b_j(o_{t+1}) \cdot \beta_{t+1}(j)}{\sum_{i=1}^N \sum_{j=1}^N \alpha_t(i) a_{ij} \cdot b_j(o_{t+1}) \cdot \beta_{t+1}(j)} \end{aligned}$$

Here, $\xi_t(i, j)$ is the probability of being in state i at time t and in state j at time $t + 1$. $\beta_{t+1}(j)$ represents the probability of a partial observation sequence from time t to $t + 1$.

HMM can be used to predict an attacker's intentions from observed alerts, enabling proactive responses to limit damage in the network segment under attack. In some studies, efforts have been made to identify similar attack intentions across different actions. For example, traditional intrusion detection systems (IDS) may fail to detect phishing attacks, DDoS attacks, and other complex network threats because they lack the technology to correlate all alert events[4]. HMM can analyze various attack patterns, correlate all events in the alert, and provide probability values in the transition process, effectively detecting complex network attack behaviors. By utilizing multiple monitoring systems and observing event names, source IP, source port, destination IP, destination port, time, and other data, HMM enables a correlation analysis of different anomalous events. This allows for sequence detection and classification of attacks using training data within the HMM model.

HMM's application in other fields also demonstrates its strong capabilities. For instance, studies have used HMM to reveal the most probable hidden stage sequences in clinical mastitis symptoms in cattle [7]. Website fingerprinting attacks can reveal the recipient in anonymous networks, posing a privacy threat to users [19]. Researchers proposed a Profiling Hidden Markov Model (PHMM) for website modeling, which effectively identifies hyperlink

transitions users might make when fingerprinting a target website [20]. Additionally, a study by William and others addressed changes in prepaid card fraud sequences, which could trigger false alarms, by updating and comparing merchant terminal HMMs. This enabled the system to dynamically create and update transaction models and automatically detect fraud in prepaid cards.

These application examples illustrate that HMM has broad potential, not only in network security but also in many other areas requiring the analysis of sequential data. Through the application of HMM, it is possible to predict and prevent potential threats more effectively, providing robust support for research and practice in related fields.

3. Research Methodology

In recent years, targeted attacks have increasingly converged around a sophisticated blend of ransomware and advanced persistent threats (APT) , marking a shift toward highly complex and covert strategies. Attackers now frequently employ advanced techniques such as code obfuscation, fileless attacks, and modifications to native operating system programs, making these three tactics among the most prevalent and concerning in modern cyber threats[22].

These attacks go beyond simple file encryption for ransom. Attackers are escalating their strategies by exfiltrating sensitive organizational data, leveraging it to intensify their demands and apply greater pressure on their victims. This dual-threat model—combining ransomware’s immediate impact with APT’s prolonged penetration—underscores a new level of threat sophistication. Targeted attacks now exhibit not only high destructiveness but also an expanded capacity for deep infiltration, thorough reconnaissance, and sustained disruption, posing significant challenges to organizational security.

This study’s innovative approach combines Elastic Net feature selection with the Hidden Markov Model (HMM) to achieve comprehensive analysis and precise risk prediction for multi-target attack behaviors. By employing Elastic Net as a feature selection tool, the model effectively identifies key features associated with attack behaviors within a high-dimensional feature space, significantly enhancing detection capabilities. When integrated with HMM, the model dynamically tracks hidden attack behaviors and, by analyzing the temporal sequence of these behaviors, offers precise predictions of potential attacks. This combination not only improves the accuracy of attack risk predictions but also enables real-time dynamic risk assessments, providing a basis for security teams to respond promptly. As a result, organizations can implement appropriate defenses before attacks escalate, minimizing potential losses and strengthening proactive security measures.

This study leverages the Hidden Markov Model (HMM) to detect abnormal attack behaviors, creating a systematic and dynamic attack detection framework. This framework encompasses data collection from multiple log sources, feature extraction, data preprocessing, HMM training and optimization, and ultimately real-time risk assessment. Through empirical risk calculations and ongoing optimization, the detection system continually learns and enhances its model performance. Below is an outline of the detection model developed in this study:

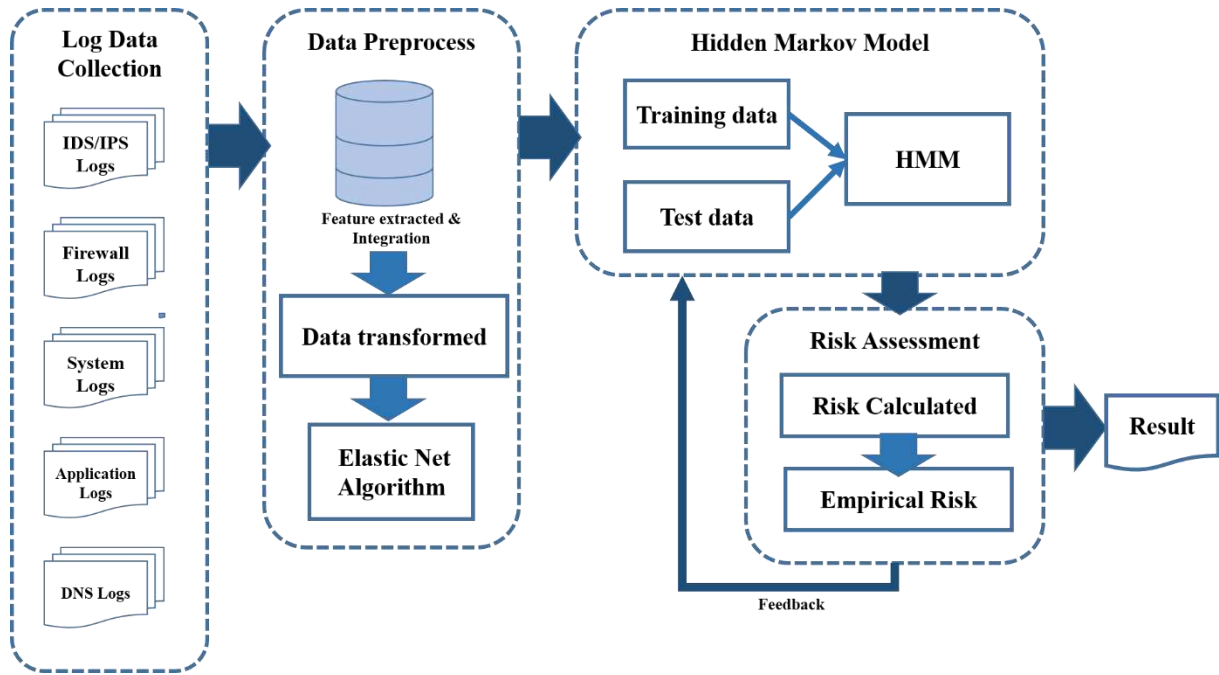


Figure 1. Research Framework

Stage 1: Data Collection from Various Logs

Data is collected from a range of network log sources (such as IDS, IPS, firewalls, etc.), covering both internal and external network logs. This comprehensive data provides richer context, facilitating more accurate identification of attack patterns.

Stage 2: Data Preprocessing

1. Feature Extraction & Integration: This step transforms raw data into usable features, ensuring the data is appropriately stored and ready for subsequent stages.
2. Data Transformation: Ensures consistency in data format, making it suitable for the next stage of model training.
3. Elastic Net Algorithm: Elastic Net is applied for feature selection and standardization, helping to identify features highly relevant to network attacks and preventing overfitting.

Stage 3: Hidden Markov Model (HMM)

HMM is used to model network behavior, and training is conducted using both training and testing data. HMM effectively captures the temporal sequence of network activities, allowing for the identification of potential attack behaviors.

Stage 4: Risk Assessment

Based on HMM predictions, the model generates specific risk values and detection outcomes. By continually incorporating new data and recalculating risk, it dynamically assesses the current network status. This ensures the model adapts to changes in the network environment in real-time and enhances its ability to detect previously unknown attack behaviors.

Stage 5: Empirical Risk Calculation and Feedback

Empirical risk is calculated to measure prediction errors on training and testing data. The results feed back into the HMM model, adjusting its parameters to further improve detection accuracy.

Through this integrated, adaptive framework, the system enables real-time responses and proactive defenses, allowing security teams to mitigate threats before they escalate.

3.1 Using Elastic Net to Identify Targeted Attack Features

In this study, to select the most discriminative features from a large volume of raw data and reduce the risk of overfitting when working with high-dimensional features, regularization regression methods were used. These methods not only train the machine learning model but also perform feature selection simultaneously. A key advantage is their ability to detect interactions between variables, with fast execution and higher precision.

Regularization regression methods are primarily categorized into three types: Lasso regression (L1 regularization), Ridge regression (L2 regularization), and Elastic Net (a combination of L1 and L2 regularization). This study finds Elastic Net to be the most suitable approach for network attack feature selection, based on the following reasons:

- (1) Feature Relevance: In network attack detection, not all features are equally important. Elastic Net retains Lasso's feature selection capability, helping identify the most critical indicators of an attack.
- (2) Feature Correlation: There may be correlations among features associated with network attacks. Elastic Net combines the strengths of Ridge, optimizing detection results by handling multicollinearity among features effectively.
- (3) Flexibility: Elastic Net allows for a balance between L1 and L2 regularization, providing flexibility to adjust according to specific attack types and data characteristics.

By leveraging these advantages, Elastic Net enhances the model's ability to isolate essential attack features with improved accuracy and resilience, supporting more precise network security analysis.

In this study, attack events that have already triggered system alerts are used as training data, encompassing all features potentially related to network attacks to form a feature matrix X and a target variable y labeled as an attack occurrence. All data is standardized. Elastic Net combines the advantages of L1 and L2 regularization, with its objective function defined as follows:

$$L(\beta) = \frac{1}{2n} \sum_{i=1}^n (y_i - X_i\beta)^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2, \quad \lambda_1, \lambda_2 \in \mathbb{R}$$

where y_i represents the actual observed values, $X_i\beta$ represents the predicted values, and n is the sample size. The term $\lambda_1 \sum_{j=1}^p |\beta_j|$ is the L1 (Lasso) component, and $\lambda_2 \sum_{j=1}^p \beta_j^2$ is the L2 (Ridge) component. Here, $\lambda_1 = \lambda\alpha$ and $\lambda_2 = \lambda(1 - \alpha)$ with λ as the regularization

strength parameter that controls the influence of regularization terms; a simpler model will have a more pronounced regularization effect. The parameter α adjusts the weight of regularization, where $\alpha = 1$ reduces Elastic Net to Lasso, and $\alpha = 0$ reduces it to Ridge. In this study, cross-validation is used to select the optimal value for α . Given the large volume of data in this study, implementing Elastic Net for feature selection poses significant challenges. To ensure that the data is effectively utilized in the model and that accurate results are achieved, data preprocessing is essential and is performed as follows:

Data Filtering and Missing Value Treatment: This study filters out and appropriately imputes missing values to maintain data integrity.

Encoding of Categorical Data: Elastic Net requires numerical inputs, so non-numerical data is converted accordingly. For instance, source IP and destination IP are encoded into numerical values; browsing risk is categorized into five levels; host information is classified based on operating system and software types, and the message field is given nine predefined levels based on content.

Clustering of Redundant Data: Highly repetitive data is grouped separately to reduce variable redundancy.

Normalization of Numerical Data: Features with widely varying ranges, such as traffic size, packet count, and severity, are normalized based on their range and characteristic attributes.

Feature Correlation Analysis: With tens of millions of records and numerous features within each time window, this study identifies and removes highly correlated, redundant features to save computation time in generating the feature correlation matrix.

These preprocessing steps ensure the data is refined and optimized for accurate and efficient application within the Elastic Net model.

The data for this study is sourced from real network observation logs, segmented into tens of millions of time windows, with risk values calculated for each window using the properties of the Markov Model. These time windows effectively capture the dynamics of multi-target attacks over various periods, providing insights into attack progression trends. By observing the data layers, the study captures fluctuations in risk values, supporting dynamic risk management with robust evaluation outcomes. This study defines and analyzes five core observable features relevant to multi-target attack behaviors:

Feature 1: Multi-probing – Abnormal fluctuations in the frequency of internal and external network connections indicate scanning and probing activities targeting network infrastructure, often a precursor to potential attacks.

Feature 2: Malicious mail – Monitoring emails containing malicious executables and Office macros, which are frequently used by attackers to penetrate internal networks. Indicators include links directing to malicious or known phishing websites, suspicious instructions (such as requests for sensitive information, link clicks, or attachment openings), and language patterns that may be auto-generated or poorly translated.

Feature 3: Browsing high-risk websites – User behavior involving visits to high-risk websites, which are susceptible to phishing attacks or attacks hiding malicious code, substantially increases the organization's exposure to threats.

Feature 4: Remote control – Attack activities initiated through remote control tools are typical in later attack stages, where attackers attempt to seize further control over the victim's system. Frequent remote operations, especially unauthorized connections, often indicate attempts to compromise internal systems. Observed characteristics include continuous and frequent communication with external IPs, use of non-standard protocols, high communication volume outside of working hours, and distinct C&C (Command & Control) signatures.

Feature 5: Data exfiltration – Monitoring unusual data transfers to external networks, a critical step in data leakage attacks, especially involving sensitive information, which poses major security risks. Characteristics include large data transfers to external IPs within short timeframes, significant data transmission during non-working hours or anomalous periods, and quick, repeated access to multiple databases or folders.

These features not only represent different stages of attack patterns but also provide a robust data foundation for accurately calculating risk values and setting detection thresholds. Ultimately, this enables effective monitoring and assessment of organizational attack risks.

3.2 Feature Weighting and Fit Assessment

This study focuses on identifying suspicious feature sets associated with targeted attacks, ultimately selecting five core features: Multi-probing, Malicious mail, Browse high-risk websites, Remote control, and Steal information. These features are derived through the combination of various conditional attributes, capturing key behaviors across different stages of an attack. To more accurately assess the significance of these features in targeted attacks, this study calculates the weight of each feature.

During data processing, we utilized 10 months of collected network behavior data, dividing it into 70% training data and 30% testing data to ensure the model's generalization capability. Within this 11-month period, the system recorded multiple attack events and issued related alerts, which serve as a benchmark for comparison. Based on this benchmark, we derived risk threshold values for each feature set, reflecting both the frequency of each feature's occurrence and its association with actual attack events.

The training dataset encompasses over seven months of network behavior data, enabling an in-depth analysis of the performance of each feature set under real attack conditions. This study employs a conditional probability model to explore the likelihood of targeted attacks occurring given the presence of specific feature sets. Based on Bayesian Inference, the model aims to calculate the probability of an attack event given a particular feature set. Through this conditional probability model, we determine the probability of each feature set's occurrence when an attack is known to have taken place. The specific probability calculation formula is as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{\text{Number of feature } i \text{ with alert}}{\text{Number of alert}}$$

where $P(A|B)$ represents the probability of an attack (A) given the presence of a specific feature set (B); $P(A \cap B)$ is the joint probability of both the feature set and an alert occurring; $P(B|A)$ is the probability of the feature set given that an attack has occurred; $P(A)$ is the prior probability of the feature set appearing within the entire dataset., and $P(B)$ represents the overall frequency of attack events, indicating the total risk of an attack.

The probability calculations in this study enable the assessment of the association between various feature sets and actual attacks, helping to determine whether these features can accurately predict future attack behaviors. Through precise mathematical modeling, the study evaluates the risk values of each feature set and optimizes predictive performance by continuously adjusting model parameters. This approach not only enhances understanding of each feature's attack risk but also supports data-driven, real-time decision-making for organizational security.

The research objective at this stage is to quantify the relationship between features and attack events, establishing threshold values for risk assessment to achieve more accurate predictions of potential attacks. Specifically, by calculating the probability of each attribute's occurrence in attack events, the study evaluates the importance of each attribute in targeted attacks, providing a clear basis for risk assessment.

This research methodology not only provides a mathematically rigorous foundation for risk assessment but also combines conditional probability with feature weighting to establish a flexible and adaptive risk evaluation framework. This approach is well-suited to handle complex network attack scenarios, enhancing the system's responsiveness and effectiveness in dynamic threat environments.

3.3 Risk Assessment Formula for Feature Set

This study provides a detailed definition and analysis of observable features associated with multi-target attack behaviors, identifying five core features with the following risk assessment equations:

Feature 1: Multi-probing

This feature captures abnormal fluctuations in internal and external network connection frequencies, reflecting scanning and login attempts that target internal and external networks—often indicators of potential targeted attacks. This study examines two core characteristics for analysis, with the observation period divided into time windows T , $t \in T, t = 1, 2, \dots$. $S(t)$ is the number of attempts by a single external IP to connect to multiple internal endpoints within time window t ; $M(t)$ is the number of attempts by multiple external IPs to connect to a single internal endpoint within time window t . These behaviors indicate potential scanning or brute-force login attempts and warrant an elevated risk assessment. The risk evaluation equation is as follows:

$$\mathcal{R}_1(T) = \alpha \left(\frac{\sum_{t=1}^T S(t)}{T} \times F_S \right) + \beta \left(\frac{\sum_{t=1}^T M(t)}{T} \times F_M \right) + \gamma(S(t) + M(t))$$

where:

- F_S and F_M represent the regularity indicators for multiple external IPs and a single external IP's connection attempts, respectively. These values capture the frequency of connection attempts within a time window, with a maximum value of 1 and a minimum of 0.
- α and β adjust the impact of $S(t)$ and $M(t)$, with these coefficients being dynamically adjusted based on training data, allowing them to change as new data is fed back into the algorithm.
- $\gamma(S(t) + M(t))$ emphasizes the influence of connection frequency on risk, where γ is also dynamically adjusted according to data from prior time windows.

Weights are dynamically set, with an increased alert threshold in the current time window if the fluctuation over the previous two windows exceeds expectations:

$$Threshold(\mathcal{R}_1) = \theta_0 + \theta_1 \Delta y(t) + \theta_2 \Delta y(t-1)$$

This approach allows for adaptive risk evaluation, where thresholds are responsive to fluctuations in connection behavior, supporting proactive detection of anomalous multi-probing activities.

Feature 2: Malicious mail

Monitoring emails containing malicious executables and Office macros is necessary, as these emails are often primary tools used by attackers to infiltrate internal networks. Key characteristics include links directing to malicious or known phishing websites, as well as suspicious instructions in the content, such as requests for sensitive information, prompts to click on links, or open attachments. The risk assessment equation for this feature is as follows:

$$\mathcal{R}_2(T) = W \cdot \sum_{i=1}^t X(t_i)$$

$$X(t_i) = E_t(K) \cdot VP_t(SE) + \frac{\gamma}{T} (A(t), M(t), P(t))$$

where $E_t(K)$ is the proportion of emails with attachments out of the total email volume K in time window t ; $VP_t(SE)$ is the proportion of significant (high-risk) IPs among the suspicious IPs observed within time window t ; $A(t), M(t), P(t)$ are represent the number of emails within time window t containing malicious executables, malicious Office macros, and links to malicious or phishing websites, respectively.

Feature 3: Browse high-risk websites

User activity involving visits to high-risk websites significantly increases the organization's exposure to phishing attacks or attacks embedding malicious code. This study assesses the risk associated with browsing high-risk websites using a time-accumulated variable, with the following equation:

$$\mathcal{R}_3(T) = \frac{1}{T} \sum_{t=1}^T P(t)$$

$$P(t) = (1 - \rho) \cdot P(t-1) + \alpha \cdot \frac{F_{hbr}}{N} \cdot R_{max}^\beta$$

where $(1 - \rho) \cdot P(t - 1)$ represents the rate at which risk decays over time within each time window, with ρ controlling the speed of decay (where ρ ranges from 0 to 1, with larger values indicating faster decay); N is the number of browsing instances within each time window; $\alpha \cdot \frac{F_{hbr}}{N} \cdot R_{max}^\beta$ represents the accumulated risk for each time window. $\frac{F_{hbr}}{N}$ is the frequency of high-risk website visits relative to total browsing instances, and R_{max}^β is the weighted highest risk value, with β serving as an exponent that emphasizes the impact of high-risk values. If $\beta > 1$, the influence of high-risk website browsing frequency on overall risk is significantly amplified, allowing for earlier detection of potential attack exposure. This equation provides a nuanced approach to evaluating browsing behavior risk, taking into account both the frequency and severity of high-risk site visits.

Feature 4: Remote control

Attack activities involving remote control tools are typically observed in later attack stages, as attackers seek to gain further control over the victim's system. Frequent remote operations, especially unauthorized connections, often indicate attempts to seize control of internal systems, posing a high risk of severe internal data leakage or resource compromise. Observable indicators include continuous and frequent communication with external IPs, use of non-standard protocols, high volumes of communication outside of working hours, and specific Command & Control (C&C) characteristics. The risk equation for remote control activities is as follows:

$$\mathcal{R}_4(T) = \alpha[\beta \cdot O(t - 1) + O(t)] + C(t)$$

where $O(t - 1)$ is the proportion of remote connection attempts in the previous observation window, representing the historical density of remote access activity, $O(t)$ is the current proportion of remote connections, reflecting the density of remote activities in the present window, β represents the decay rate of remote connection density from the previous observation window, α is a weight factor that balances the impact of remote connection activities across both the current and previous time windows, $C(t)$ indicates connection to a C&C server, where a connection is assigned a value of 1, and no connection is assigned a value of 0.

This formula captures the persistence and frequency of remote control attempts, weighted by historical and real-time activity, as well as the presence of C&C connections. By tracking these factors, the equation provides a dynamic assessment of remote control risk, highlighting unauthorized remote activities and potential C&C communication as key indicators of advanced attack stages.

Feature 5: Steal information

This feature monitors abnormal data transmission behaviors to external networks, a critical step in data exfiltration attacks. The transmission of sensitive information, particularly large volumes over short periods, poses significant security risks. Key indicators include: substantial data transferred to external IPs within a short time, high-volume transmissions during non-working or anomalous hours, and rapid access to multiple distinct databases or folders. The risk equation for data exfiltration is as follows:

$$\mathcal{R}_5(T) = \alpha \frac{r_V(t)}{r_V(t-1)} + \beta \frac{r_P(t)}{r_P(t-1)} + \gamma \frac{1}{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 + \varepsilon} - 1$$

where $r_V(t)$ and $r_V(t-1)$ represent the current and previous data volume rates, respectively, $r_P(t)$ and $r_P(t-1)$ denote the current and previous packet volume rates, α and β are coefficients adjusted based on historical data, The final term evaluates the stability or volatility of the transmission behavior using standard deviation σ . Specifically, it calculates the inverse square of the standard deviation to amplify the impact of regular transmission patterns on the risk score. A low standard deviation indicates stable transmission behavior, suggesting that an attacker may be executing data transfers at specific intervals.

The formula leverages geometric growth rates for traffic and packet volumes, highlighting instances of rapid increase to elevate risk values in high-risk environments. This non-linear transformation of the standard deviation intensifies the risk assessment when transmission patterns exhibit strong regularity, enabling real-time detection of abrupt changes in data transfer activity. This approach is especially effective for identifying structured, low-variability exfiltration patterns characteristic of advanced data theft scenarios.

Finally, to calculate the overall risk value from the five feature sets, this study first applies Min-Max normalization to each feature set, scaling each risk value to a range of [0,1]. The resulting feature set vector is $\{\mathcal{R}_n(T)\}$, $n = 1, 2, \dots, 5$. Time series analysis is then used to evaluate changes in the risk values, with an autoregressive (AR) model applied to detect temporal dependencies and variations in the risk scores across time periods. This approach provides an effective method for dynamic risk assessment, enhancing the depth and accuracy of time series analysis in risk detection. The AR model equation is:

$$\mathcal{R}(t) = \alpha \mathcal{R}(t-1) + e(t)$$

where the parameter α is estimated to analyze the influence of the previous time window's risk value $\mathcal{R}(t-1)$ on the current window's risk $\mathcal{R}(t)$. If $\alpha > 1$, it indicates a significant increase in risk. Next, differencing is used for regression analysis to capture the trend of risk values over time:

$$\Delta \mathcal{R}(t) = \mathcal{R}(t) - \mathcal{R}(t-1)$$

The risk value changes across different time periods are weighted according to the importance of each period or its specific network environment, yielding the following risk value equation:

$$\mathcal{R}_{all} = \alpha_1 \Delta \mathcal{R}(t) + \alpha_2 \Delta \mathcal{R}(t-1)$$

The feature sets in each time window, represented by the above risk values, form the hidden layer in the Hidden Markov Model (HMM). By inputting these values into the HMM, a dynamic risk assessment for each stage is achieved, allowing for real-time evaluation of the current risk level.

3.4 Markov Model Detection Model

The Hidden Markov Model (HMM) is a statistical model based on hidden states, widely used in time series analysis. In this study, HMM is applied to network attack detection,

modeling the dynamic changes in network behavior to identify potential malicious attack activities. As an extension of the Markov Model, HMM includes both unobservable hidden states and observable sequences, making it particularly effective in handling complex, indirectly observable system dynamics.

In HMM, the hidden states form a Markov chain where each transition depends only on the current state and not on all previous states, adhering to the Markov property of "memorylessness." Specifically, state transitions follow a first-order Markov process, where each state transition depends solely on the preceding state. However, the model can be further extended to an n-order Markov process, where each transition depends on the previous n states, thereby increasing the system's complexity and accuracy.

This dual structure of hidden states and observable sequences allows HMM to capture intricate patterns in network behavior, providing a robust framework for detecting subtle indicators of potential attacks in complex environments.

In this study, the HMM model utilizes fields such as login time, source IP, source port, destination IP, destination port, traffic, packet count, message, website risk, email file, protocol, port number, remote behavior, and FTP activity as the observation layer. From these, five feature sets—Multi-probing, Malicious mail, Browse high-risk websites, Remote control, and Steal information—are derived to form the hidden layer, which is then used to classify behaviors into normal, suspected attack, or confirmed attack states.

The core of HMM lies in the dynamic transitions between hidden states. Although these states are not directly observable, they manifest through shifts in the observable sequences. In the HMM, network behaviors are treated as observable sequences, while potential attack behaviors are hidden within the state transitions. Training the HMM model allows it to learn the patterns of these state transitions, effectively identifying potential anomalies or attack behaviors within the network. This flexibility and interpretability make HMM advantageous for dynamic behavior modeling and anomaly detection. The basic components of the HMM in this study are as follows:

The statistical model is designed with hidden state sets $S = \{s_1, s_2, \dots, s_t\}$ over t time windows, representing "normal behavior", "suspected attack behavior", and "confirmed attack behavior".

The observed states $O = \{o_1, o_2, \dots, o_m\}$ represent the extracted features.

The initial state distribution is $\pi = \{\pi_i\}$, where $\pi_i = P(s_i)$ denotes the probability of the system being in a particular state at the initial time.

The state transition matrix $A = \{a_{ij}\}$ contains elements $a_{ij} = P(s_{t+1} = j | s_t = i)$, representing the transition probability from one state to another.

The observation matrix $B = \{b_j(k)\}$ defines the probability of observing k given the hidden state j . The transition probability over n states can be expressed as a_{ij}^n , giving $b_j(k) = P(O_t = k | s_t = j)$.

After identifying key attack features, this study applies the Hidden Markov Model (HMM)

for iterative training, enabling dynamic adaptation to and learning of network behavior variations. By updating model parameters within each time window, HMM continually optimizes over time, improving the accuracy of attack behavior identification and prediction. This time-window-based dynamic evaluation mechanism allows the model to more accurately detect potential threats and enhances risk assessment accuracy. Consequently, the model can trigger real-time defense measures to reduce potential damage to the organization from attacks.

This study calculates threshold values for three risk stages—low risk (normal behavior), medium risk (suspected attack behavior), and high risk (confirmed attack behavior)—based on the identified attributes:

- **Low Risk:** Defined as a state where all behavior features in the feature matrix fall within the historical average range, showing no significant anomalies or attack patterns.
- **Medium Risk:** Defined as the stage when certain features exceed normal ranges but do not yet indicate full-blown attack behavior. This may include unusual but non-destructive actions, such as an increase in abnormal traffic or multiple failed login attempts.
- **High Risk:** Marked as high risk when critical features—such as abnormal network traffic, frequent connection attempts, or unusual access to sensitive internal data—reach or exceed set attack thresholds. At this point, it can be confirmed that an attack is in progress, and immediate defense measures should be initiated.

Observations in this study indicate that attack behaviors do not unfold in a set sequence but rather exhibit nonlinear and irregular characteristics depending on the attacker's strategy and environmental changes. Therefore, HMM, as a modeling tool for capturing hidden state transitions, effectively adapts to the randomness and unpredictability of attack behaviors. This makes HMM particularly suitable for dynamically detecting and assessing varying levels of threat within complex network environments.

3.5 Empirical Risk

This study applies the Hidden Markov Model (HMM) for network attack detection, using Empirical Risk as a tool for model optimization. Empirical Risk is a core concept in machine learning and statistics, typically applied in supervised learning to measure a model's prediction error on training data, which is essentially the average misclassification or regression error in the model.

Given a dataset $\{x^t, y^t\}$, where $t = 1, 2, \dots, n$. x^t represents the value of the input feature set (in this study, each data point has five feature values), and y^t is the corresponding label. The loss function $L(\theta, X^t, y^t)$ evaluates the model's prediction error. Empirical Risk is defined as:

$$R_n(\theta) = \frac{1}{n} \sum_{t=1}^n \text{Loss}(y^t - \theta \cdot X^t)$$

where θ represents the model parameters, $\{X^t, y^t\}$ represents the t -th training data, the loss function $L(\theta, X^t, y^t)$ measures the difference between the model's prediction on

sample X^t and the actual label y^t .

In this network attack detection study, Empirical Risk is used to evaluate the accuracy of the model's prediction of whether network behavior is an attack. First, network traffic logs are labeled as "normal" or "attack" to form supervised training data. After training the HMM-based attack detection model, Empirical Risk on the training data is calculated to adjust model parameters. This minimizes the risk and enhances detection accuracy, ensuring that the model is well-optimized for real-time network behavior assessment.

4. Experimental Results and Discussion

This chapter delves into the performance of the Hidden Markov Model (HMM) in detecting targeted attack behaviors. Modern network attacks, particularly targeted or advanced persistent threats (APTs), often display characteristics of stealth and cyclical patterns. These characteristics necessitate the design of detection models that account for sampling intervals and capture subtle behavioral shifts over time. Our experiment employs a rolling sampling period, which enhances the model's capacity to detect evolving network behaviors effectively. This rolling approach captures fluctuations in risk values associated with cyclical attack behaviors, enabling earlier identification of sustained attack actions.

Experimental data show that this detection model accurately identifies 94.7% of attacks, including early detection of threats overlooked by the organization's internal monitoring systems. The model analyzes the risk variations across different time intervals for each IP, allowing for precise tracking and adaptive risk assessment. Through this approach, the model enhances detection accuracy, providing a more robust and proactive defense against sophisticated and stealthy network threats.

4.1 Data Preprocessing and Feature Extraction

In this study, data related to targeted attacks are collected from network log sources across multiple monitoring systems (such as IDS, IPS, WAF, and traffic logs). To select the most discriminative features from the vast array of raw log attributes and to prevent overfitting when handling numerous features, Elastic Net is employed to identify targeted attack characteristics. Key features are extracted from various log fields, including time, source IP, source port, destination IP, destination port, traffic, packet count, message, website risk, email file, protocol, port number, and remote connection information. Each sample is labeled as "normal" or "attack behavior," using IP as a unique key.

The data is segmented into multiple time windows to allow more granular analysis of activities within each interval. All features are compiled into a feature matrix X and a corresponding target vector y , where y contains the labels for targeted attacks. All potential network behaviors (such as connection frequency, IP changes, scanning activity, and abnormal logins) are converted into numerical feature variables. Given that targeted attack behaviors are often multidimensional and highly correlated, these features form a high-dimensional matrix.

Using the Elastic Net model for feature selection assigns weights within the feature space, penalizing less important features by shrinking their coefficients, thereby retaining only the most representative attack features. For instance, retained features might include the number of connections to malicious IPs, suspicious file operations, or anomalous network traffic within specific time periods. These selected features enable the identification of potential attack patterns, providing a robust basis for detecting targeted threats.

Below is a list of features selected by the Elastic Net model, along with their regression coefficients. The regression coefficient represents the weight of each feature, with higher absolute values indicating greater discriminative power in identifying targeted attack behaviors. The larger the absolute value of the weight, the more significant the feature is in distinguishing targeted attack activities.

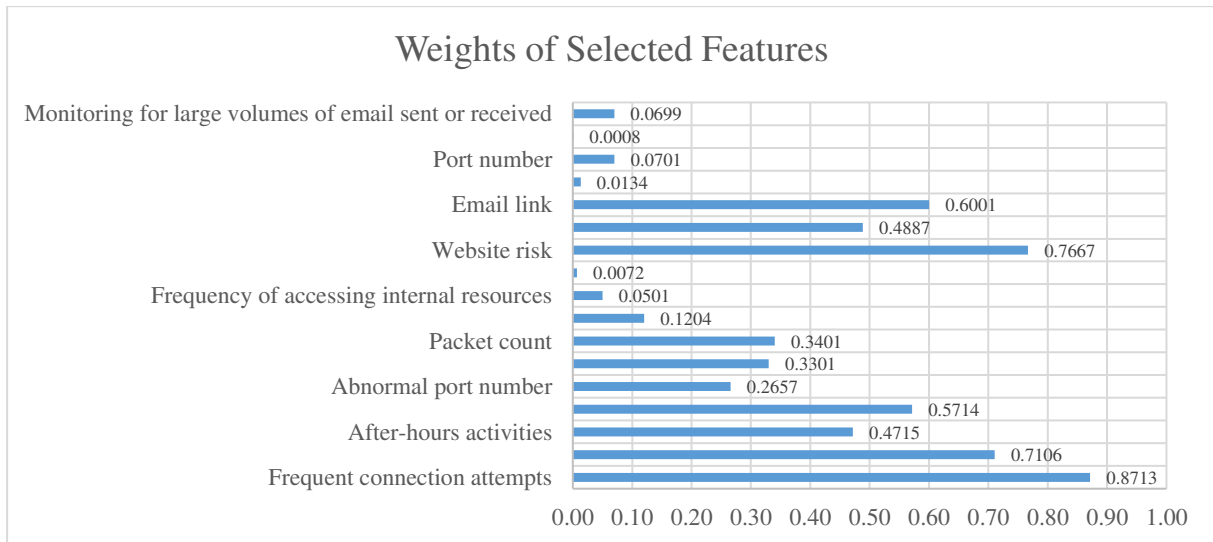


Figure 2. Weights of Selected Features

In this study, real attack data is used to analyze features of both attack and non-attack behaviors within historical data, and statistical methods are applied to quantify threshold values for risk levels. This approach aims to identify potential threats in the early stages of an attack. When the system detects that the overall risk score of all features reaches the medium- or high-risk threshold, an alert mechanism is triggered to notify security personnel, prompting intervention and protective measures. This process significantly enhances network security and reduces potential damage by preventing the attack from escalating further.

4.2 Risk Value Calculation and Prediction

After determining the key features and selecting an appropriate time window size, this study uses the Hidden Markov Model (HMM) to calculate the risk level within each time window and predict potential future targeted attack behaviors.

For overall risk assessment, a Confusion Matrix is used as an evaluation metric to assess the effectiveness of attack detection in each time window. High-risk data, as identified by HMM, is considered indicative of attack activity, and the assessment is compared with attack alerts issued by the organization. Initially, detection performance on training data was suboptimal, primarily due to the high sensitivity of the detection system, which resulted in a high number of Type II errors (false positives), misclassifying non-attack events as attacks.

Through iterative adjustments via a feedback mechanism, the detection performance improved significantly after two months of training, eventually achieving over 90% accuracy after three months. This improvement underscores the efficacy of feedback in refining the training data and enhancing the model's detection capability.

In the subsequent phase, two months of test data were used to evaluate the trained model, achieving an average detection rate of 94.71%, as expected. Beyond identifying attacks that the organization's detection systems had already flagged, this study's primary contribution is its ability to detect attacks early. To validate the effectiveness of HMM in targeted attack prediction, real network attack data were used in experiments. The results show that HMM can significantly improve network security monitoring accuracy, enabling early identification of

high-risk targets before the system issues an alert. Below is a description of six confirmed attack cases that were detected early by this study:

For IP1, initial data indicated a high risk score, which later decreased to a medium-risk level. This change is attributed to an early-stage remote connection that initially assigned a higher risk score; however, as remote connection activity did not persist, the risk score declined. Other suspicious activities kept the risk at a medium-high level. In this case, the model identified the attack in a medium-high risk state 1–2 days before the organization's IDS system, primarily due to abnormal remote connection detection.

Notably, the cases of IP2 and IP6 are significant. These two cases consistently showed medium-low risk in the detection system, while this study flagged them as suspicious. In a real-world detection system, these attacks were challenging to detect due to the lack of pronounced anomalous behavior. According to this study's criteria, however, these attacks could have been detected 9 days earlier than by the real detection system.

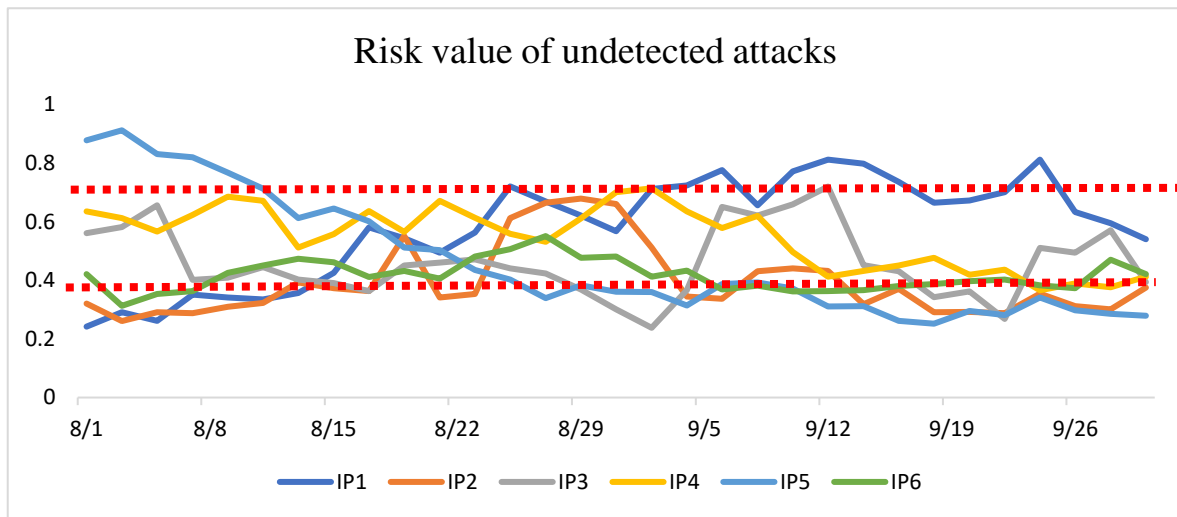


Figure 3. Risk Variation Before Detection of Attack

The experimental results confirm that the HMM model can be effectively applied for real-time network behavior monitoring. When abnormal behavior is detected, the model automatically triggers alerts and protective measures, enabling early detection of attack activities. This capability not only reduces the risk of successful attacks but also significantly minimizes potential losses to the organization.

5. Conclusion

In today's environment of rapid data growth, cybersecurity analysts face unprecedented challenges, making the selection of effective analytical methods a critical factor in data analysis. This study leverages the Hidden Markov Model (HMM) as a tool for analyzing targeted attack sequences, effectively uncovering hidden attack patterns and identifying anomalous penetration techniques. Initially, Elastic Net was applied to select critical features associated with targeted attacks. Combining L1 and L2 regularization, Elastic Net balances model complexity and accuracy, helping this study to identify key variables while filtering out noise, thus accurately capturing concealed anomalies in targeted attacks.

Next, the study calculated the initial probabilities for each risk state based on the training dataset and integrated them into the HMM, enhancing computational efficiency and predictive accuracy. Experimental results demonstrate that this approach effectively detects atypical targeted attack methods, providing precise predictions and real-time risk assessment. This contributes to optimizing security strategies and bolstering organizational defense capabilities.

Our experiments confirm that HMM is well-suited for handling dynamic, time-dependent attack behaviors, especially when normal behaviors exhibit regular periodicity, allowing HMM to model these temporal dependencies effectively, increasing accuracy, and reducing false positives. To further address data imbalance, future work could incorporate data resampling techniques or cost-sensitive learning methods to improve detection of minority classes, such as malicious attack samples.

While HMM performs well in prediction, handling increasingly complex attack patterns may challenge its accuracy. This study foresees the integration of other machine learning techniques, such as deep learning, to further enhance predictive capabilities, extending the model's effectiveness in combating sophisticated threats.

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